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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

AC3102-1- INFORMATION THEORY AND CODING
(For ACT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- For a source with an alphabet of 8 equally likely symbols, what is the entropy?
A) 0.125 bits/symbol
B) 3 bits/symbol
C) 8 bits/symbol
D) $\log_2(8!)$ bits/symbol
- The Source Coding Theorem states that the average codeword length (L_{avg}) for a source code can be made arbitrarily close to:
A) The number of symbols in the alphabet.
B) The maximum information content of any symbol.
C) The source's entropy (H)
D) The probability of the most frequent symbol
- What is the primary advantage of Lempel-Ziv (LZ) algorithms over Huffman coding
A) They always produce a shorter average codeword length
B) They are a type of fixed-length code
C) They do not require prior knowledge of the source's symbol probabilities
D) They are specifically designed for images and video
- What does the term "zero-memory source" imply?
A) The source has no storage
B) The probability of a symbol depends on the previous symbol
C) The symbols are generated independently of each other
D) The source only generates two symbols
- What is the information content of a message with a probability of 0.5?
A) 0.5 bits
B) 1 bit
C) 2 bits
D) 0 bits
- What is the relationship between probability and information content?
A) Directly proportional
B) Inversely proportional
C) They are unrelated
D) They are equal
- The entropy of a source is maximized when:
A) The symbols are equally likely
B) One symbol has a probability of 1
C) The probabilities are skewed
D) The source alphabet is very large
- The term "discrete" in Discrete Memoryless Source (DMS) refers to the source's ability to produce:
A) A continuous range of symbols
B) Symbols at discrete time intervals
C) An infinite number of symbols
D) Symbols from a finite alphabet
- What is the dimension of an H^T matrix of an (n, k) LBC?
A) $k \times n$
B) $n \times k$
C) $(n-k) \times n$
D) $n \times (n-k)$

10. The generator polynomial for a (7, 4) binary cyclic code is given by $g(x) = 1 + x + x^3$. The parity for the message (1 0 0 1)
- A) 111 B) 010
C) 011 D) 110
11. Consider the (15, 11) cyclic code generator by $g(x) = 1 + x + x^4$. Parity bits for the message 11001101011 is
- A) 1111 B) 0000
C) 1010 D) 0111
12. Number of shift registers used in the syndrome calculating circuit of a (15, 5) cyclic code is
- A) 5 B) 10
C) 15 D) Insufficient Information
13. Consider a Rate 1/5 repetition code. What is the error-correcting capability of this code?
- A) 1 B) 2
C) 3 D) 4
14. The first entry of each row of the standard array for an (n, k) linear block code is called as
- A) Coset Leader B) Coset Member
C) Parity D) Code Vector
15. If we encode the following information sequence using $R=1/3$ repetition code, what will be the coded sequence? Information sequence: 1101
- A) 001001000001 B) 000111000111
C) 111111000111 D) 111000111000
16. Which of the following polynomial is a valid $g(x)$ for the construction of a (7,4) cyclic code?
- A) $1+x^2+x^3$ B) $1+x+x^2$
C) $1+x+x^2+x^3+x^4$ D) $x+x^3$
17. The maximum number of encoder outputs that can be affected by a single input is called as
- A) Column distance function B) Code length
C) Code rate D) Constraint length
18. The number of stages required in the Trellis diagram of a (3, 1, 2) systematic convolution code for an input of length 8 is
- A) 8 B) 11
C) 5 D) 10
19. Code rate of a convolution code is 2/3. The number of input lines to the encoder is
- A) 3 B) 2
C) 1 D) 4
20. For a (3, 1, 2) systematic convolution code, the number of '0's to be inserted after the input bits are exhausted to obtain the overall code vector is
- A) 2 B) 3
C) 1 D) Insufficient information

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Morse code uses a sequence of dots and dashes to transmit information. The dash is represented by a pulse of 3 ms duration and a dot of 1 ms duration. The probability of a dash is one-third that of a dot. Consider that a 1 ms duration of gap is given in between the symbols.
Calculate
- Self-information of a dot and a dash.
 - Average information content of a dot-dash code
 - Average rate of information.

Marks BT* CO* PO*

6 L*3 1 1

- b) Consider a zero-memory source emitting three symbols X, Y and Z with respective probabilities $\{0.6, 0.3, 0.1\}$. Calculate
- Entropy of the source.
 - All symbols and the corresponding probabilities of the second-order extension of the source. Find the entropy of the second-order extension of the source.
 - Show that $H(S^2) = 2H(S)$.
- c) Construct the operation table for GF (5).
- | | | | |
|---|----|---|---|
| 6 | L3 | 1 | 1 |
| 4 | L3 | 2 | 1 |
2. a) Consider a source with
 $S = (A, B, C, D, E)$
 $P = (0.4, 0.25, 0.15, 0.12, 0.08)$.
Construct the binary codewords for the symbols using Shannon's algorithm.
- | | | | |
|---|----|---|---|
| 6 | L3 | 1 | 1 |
|---|----|---|---|
- b) Design a system to report the heading of a collection of 400 cars. The heading levels are heading straight (S), turning left (L) and turning right (R). This information is to be transmitted every second. Construct a model based on the following test data:
- On an average, during the reporting interval, 200 cars were heading straight, 100 were turning left and the remaining were turning right.
 - Out of 200 cars that reported heading straight, 100 reported going straight during the next reporting period, 50 turning left and the remaining turning right during the next period.
 - Out of 100 cars that reported as turning during a signaling period, 50 continued the turn and the remaining headed straight during next signaling interval.
 - The dynamics of the cars did not allow them to change their heading from left to right or right to left during subsequent reporting periods.
- Find the entropies of each state, the source entropy and the rate of information.
- | | | | |
|---|----|---|---|
| 6 | L3 | 1 | 1 |
|---|----|---|---|
- c) Write the condition under which a set F with binary operations addition and multiplication is called as a field.
- | | | | |
|---|----|---|---|
| 4 | L3 | 2 | 1 |
|---|----|---|---|
3. a) A source emits one of the following 8 symbols with the following probabilities
 $P \{0.40, 0.25, 0.12, 0.08, 0.05, 0.05, 0.03, 0.02\}$.
Construct a binary code using the Huffman encoding procedure and find its efficiency.
- | | | | |
|---|----|---|---|
| 6 | L3 | 1 | 1 |
|---|----|---|---|
- b) Prove that $(x^2 + x + 1)$ is an irreducible polynomial over $F_2[x]$
- | | | | |
|---|----|---|---|
| 6 | L3 | 2 | 1 |
|---|----|---|---|
- c) Show that the set of integers under addition $(Z, +)$ is a group
- | | | | |
|---|----|---|---|
| 4 | L3 | 2 | 1 |
|---|----|---|---|

Unit – II

4. a) The Parity check bits of a (8,4) linear block code are given by
- $$c_5 = d_1 + d_2 + d_4$$
- $$c_6 = d_1 + d_2 + d_3$$
- $$c_7 = d_1 + d_3 + d_4$$
- $$c_8 = d_2 + d_3 + d_4$$
- where d_1, d_2, d_3, d_4 are data bits
- Find the generator and parity check matrices
 - Find all the possible code vectors
 - Find the minimum hamming weight
- | | | | |
|----|----|---|---|
| 10 | L3 | 3 | 1 |
|----|----|---|---|

- b) Design a single error correcting code with a message block size of 11.

6 L3 3

5. a) Consider a (15,11) cyclic code generated by $g(x) = 1 + x + x^4$. Devise a feedback shift register encoder circuit. Illustrate the encoding procedure with the message vector 10010110111 by listing the state of the registers.

10 L3 4 1

- b) For a (7, 4) cyclic code, the received vector $Z(x)$ is 1110101 and the generator polynomial is $g(x) = 1 + x + x^3$. Draw the syndrome calculation circuit and correct the single error in the received vector.

6 L3 3 1

6. a) Consider a (15,5) binary cyclic code with $g(x) = 1 + x + x^2 + x^4 + x^5 + x^8 + x^{10}$

i. Draw the encoder circuit

ii. Find the codeword for the data $d(x) = 1 + x^3 + x^4$

iii. Draw the syndrome calculating circuit.

10 L3 4 1

- b) For a systematic (7, 4) linear block code, the parity matrix P is given by

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

A single error has occurred in each of these received vectors. Detect and correct those errors.

(a) RA = [0111110]

(b) RB = [1011100]

(c) RC = [1010000]

6 L3 4 1

Unit – III

7. a) Consider a (2,1,2) convolutional code with generator sequences $g_1 = (1 \ 1 \ 1)$ and $g_2 = (1 \ 0 \ 1)$. Draw its Trellis Diagram. Hence, decode the sequence {11, 11, 10, 01, 00, 10, 11} using the Hard Decision Viterbi Decoding Algorithm.

10 L3 5 1

- b) For the convolutional encoder shown in Figure Q.7(b), the information sequence is $d=10011$. Find the output sequence using time domain and transform domain approach.

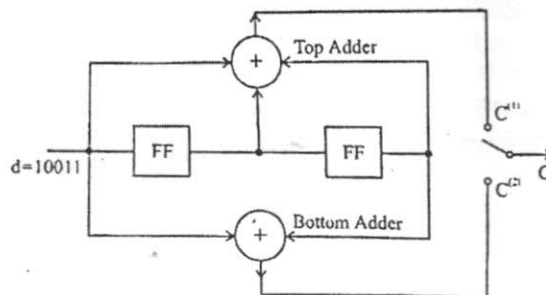


Figure Q.7 (b)

6 L3 5 1

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- a) Consider a (3, 1, 3) with generator polynomials: $g_1 = (1011)$, $g_2 = (1111)$ and $g_3 = (1101)$. Draw its encoder diagram. Also find the encoded sequence for the input (11101101) using the time-domain approach. Verify the answer using the transform-domain approach.
- b) Given a (2,1,2) binary convolutional code with a partially completed state diagram as shown in 8(b). Construct the complete state diagram and sketch the encoder diagram.

10 L3 5 1

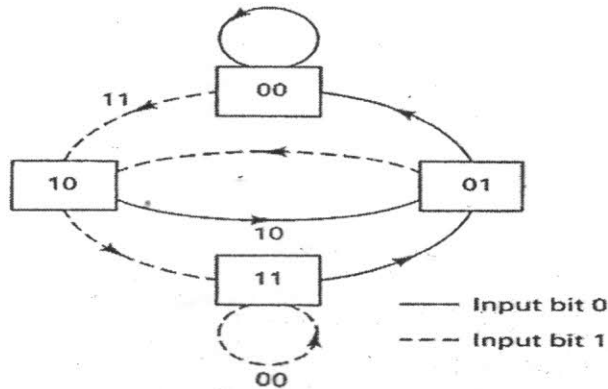


Figure Q.8 (b)

6 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
